Name: _____

TOPIC TEST 2

Chapter 4: Functions and graphs

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 7 short answer questions (57 marks)
- Part C: 2 analysis questions (14 marks)

Part A

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Which of the following is an even function?

A $f(x) = 2x^2 - 3x + 1$

B $f(x) = x^4 - 7x$

C $f(x) = x^4 - 7x^2$

D $f(x) = x^3 + 4x$

E $y = x^4 + x^3 - 3$

[1 mark]

2 For the function $f(x) = \begin{cases} x^3 & \text{if } x \geq 3 \\ 5 & \text{if } 0 \leq x < 3 \\ 2x+1 & \text{if } x < 0 \end{cases}$ evaluate $f(3) - f(2) - f(-1)$.

A 1

B 21

C -1

D 23

E 5

[1 mark]

3 Find the equation of the function that is 5 units to the left of $y = x^2$ and 1 unit down.

A $y = (x + 5)^2 - 1$

B $y = (x + 5)^2 + 1$

C $y = (x - 5)^2 + 1$

D $y = 5x^2 - 1$

E $y = (x - 5)^2 - 1$

[1 mark]

4 Find the domain and range of $y = \sqrt{2 - x}$.

A Domain $\{x: \text{all real } x\}$, range $\{y: \text{all real } y\}$

B Domain $\{x: \text{all real } x \geq 2\}$, range $\{y: \text{all real } y \geq 0\}$

C Domain $\{x: \text{all real } x \neq 2\}$, range $\{y: \text{all real } y \geq 0\}$

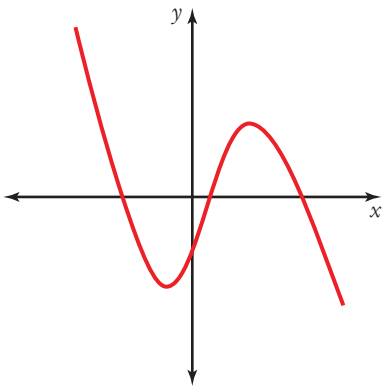
D Domain $\{x: \text{all real } x \leq 2\}$, range $\{y: \text{all real } y \geq 0\}$

E Domain $\{x: \text{all real } x \geq 2\}$, range $\{y: \text{all real } y \leq 0\}$

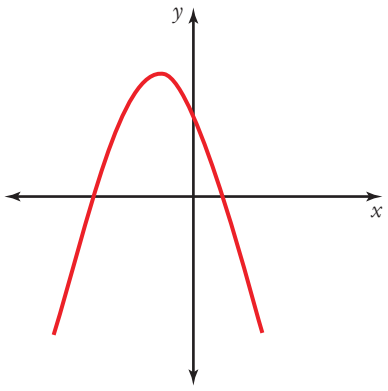
[1 mark]

5 Which graph is not a function?

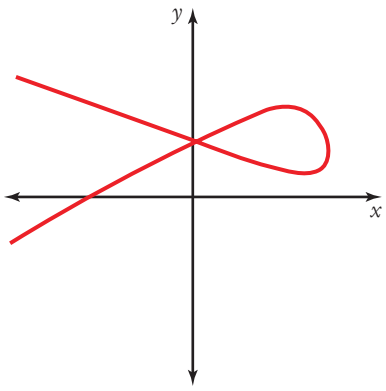
A



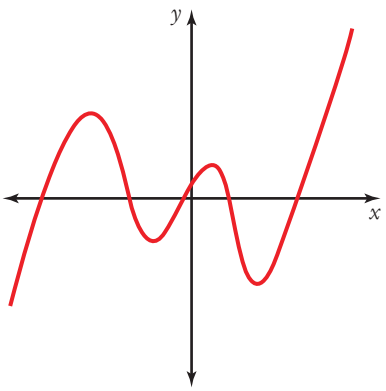
B



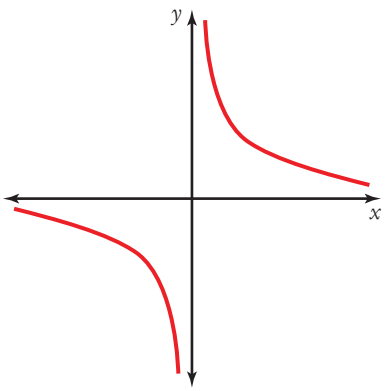
C



D



E



[1 mark]

Part B

7 short answer questions
57 marks
Show your working where appropriate.

1 Sketch the following graphs.

a $y = \dfrac{2}{x-1}$

b $x^2 - 2x + y^2 + 2y + 1 = 0$

c $f(x) = (x - 1)^3$

d $x = -3$

e $f(x) = x^2 + 3x$

[10 marks]

2 Find the gradient of each of the following lines.

a with x-intercept of 1 and y-intercept of 4

b that makes an angle of 60° with the positive x-axis (exact value)

c perpendicular to the line $x - 5y - 3 = 0$

d passing through points $(-7, 6)$ and $(4, -3)$

[9 marks]

3 Find the equations of the following.

a The line parallel to the line $y = 3x + 1$ that passes through $(-1, 6)$.

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b The quadratic function passing through (0, 1), (3, 4) and (1, −2).

c The circle with centre (2, −2) and radius 4.

[10 marks]

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4 For the quadrilateral $ABCD$ with vertices $A = (-1, 0)$, $B = (2, 4)$, $C = (7, 4)$ and $D = (4, 0)$:

a show that $ABCD$ is a rhombus

b show that the diagonals AC and BD bisect each other.

[8 marks]

5 For the function $y = -3x^2 + x - 4$:

a find the discriminant and state the number and type of its zeros

b find the equation of its axis of symmetry

c find the coordinates of its turning point and state whether it is a maximum or minimum

d sketch the graph of the function showing the axis, turning point and any intercepts.

[8 marks]

6 For $P(x) = a(x - b)(x - c)(x - d)$:

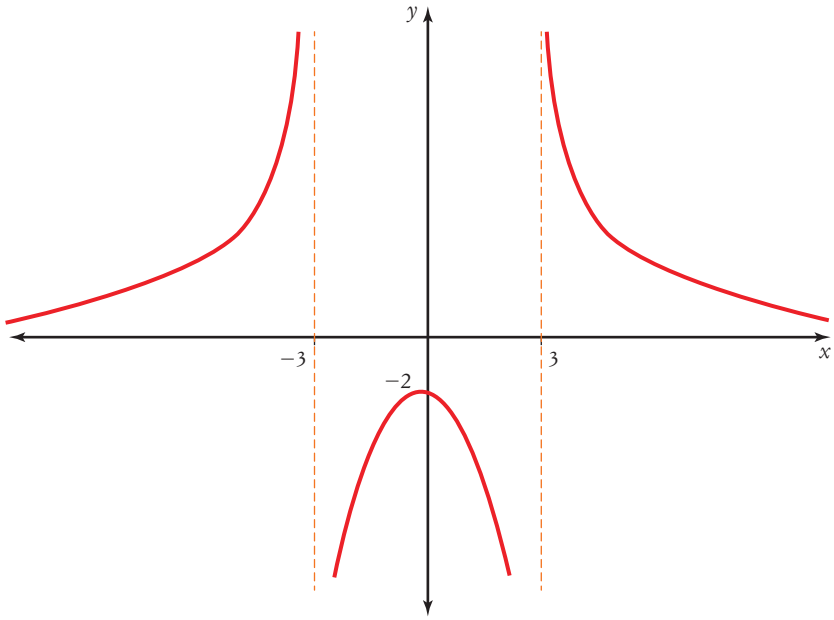
a how many zeros does the function have, and what are they?

b what is the degree of the polynomial?

c if $P(x)$ is a monic polynomial with $b = 2$, $c = -1$ and $d = 4$, sketch $P(x)$.

[6 marks]

7 For the graph below, do the following.



a State where the graph is discontinuous.

b Is the graph a function? _____

c State the domain over which the graph:

i has a positive gradient _____

ii has a negative gradient _____

iii is increasing _____

iv is decreasing. _____

[6 marks]

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Part C

2 analysis questions
14 marks
Show your working where appropriate.

1 For the relation $y^2 = 4ax$:

a state two functions that make up this relation.

_____ _____

b draw a graph showing the general shape of this relation when $a > 0$.

c show that the point $Q(aq^2, 2aq)$ lies on the graph of $y^2 = 4ax$.

d show that the gradient of line PQ, where $P = (ap^2, 2ap)$, is given by $m = \dfrac{2}{p + q}$

e show that the equation of PQ is given by $2x - (p + q)y + 2apq = 0$.

[9 marks]

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2 **a** Sketch the graphs of $y = \dfrac{2}{x-3}$ and $y = 1$ on the same set of axes.

b Use your graph to solve $\dfrac{2}{x-3} \leq 1$.

[5 marks]

Total marks: 76

Answers

Part A **[1 mark each]**

1 **A** $f(-x) = 2(-x)^2 - 3(-x) + 1 = 2x^2 + 3x + 1 \neq f(x)$

B $f(-x) = (-x)^4 - 7(-x) = x^4 + 7x \neq f(x)$

C $f(-x) = (-x)^4 - 7(-x)^2 = x^4 - 7x^2 = f(x)$, so the function is even.

D $f(-x) = (-x)^3 + 4(-x) = -x^3 - 4x \neq f(x)$

E $f(-x) = (-x)^4 + (-x)^3 - 3 = x^4 - x^3 - 3 \neq f(x)$

$\therefore$ **C** **[1 mark]**

2 $f(3) - f(2) - f(-1) = 3^3 - 5 - [2 \times (-1) + 1]$

$= 27 - 5 + 1$

$= 23$

$\therefore$ **D** **[1 mark]**

3 The function 5 units to the left of $y = x^2$ is $y = (x + 5)^2$.

The function 1 unit down from $y = x^2$ is $y = x^2 - 1$.

The function 5 units to the left of $y = x^2$ and 1 unit down is $y = (x + 5)^2 - 1$

$\therefore$ **A** **[1 mark]**

4 For $y = \sqrt{2-x}$

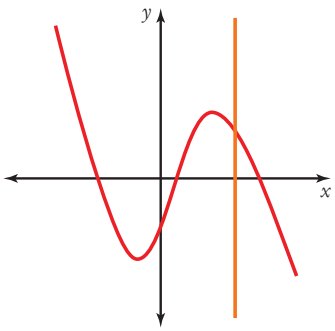
$2 - x \geq 0$

$2 \geq x$

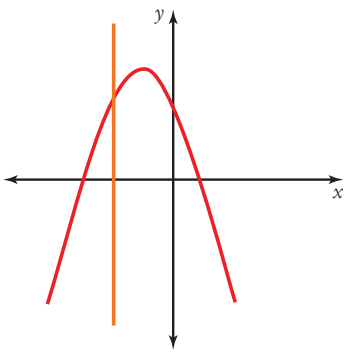
$y \geq 0$

$\therefore$ **D** **[1 mark]**

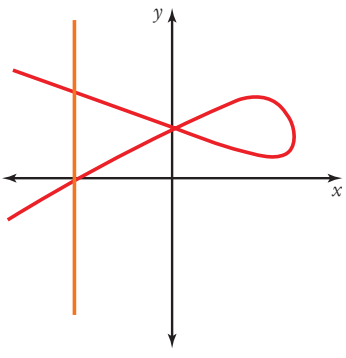
5 **A**



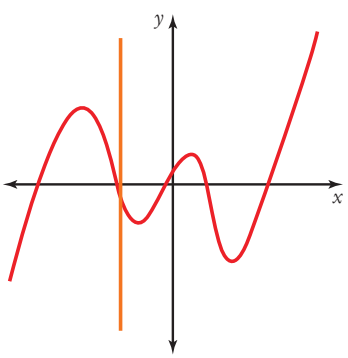
B



C

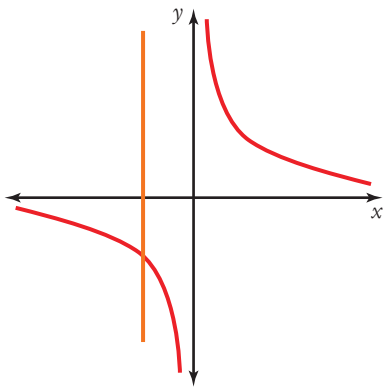


D



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E

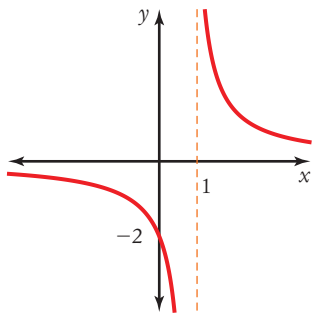


The vertical line crosses graph C in more than one place so it is not a function.

$\therefore$ **C** **[1 mark]**

Part B

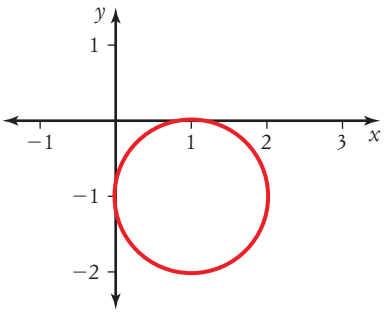
1 a This is a hyperbola in the 1st and 3rd quadrants with asymptotes at $x = 1$ and $y = 0$.



[2 marks]

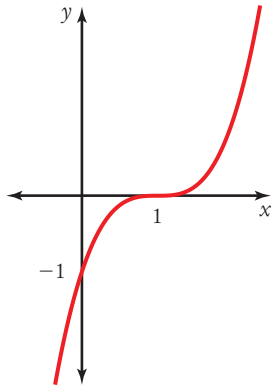
b
$$x^2 - 2x + y^2 + 2y = -1$$
$$x^2 - 2x + 1 + y^2 + 2y + 1 = -1 + 1 + 1$$
$$(x - 1)^2 + (y + 1)^2 = 1$$

This is a circle with centre $(1, -1)$ and radius 1 unit. **[2 marks]**



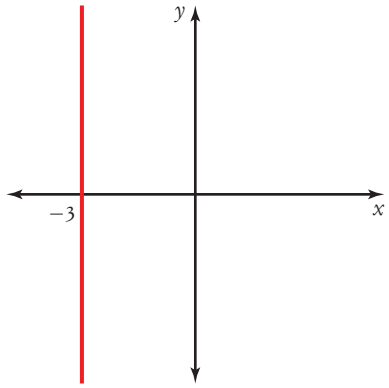
[1 mark]

c This is a cubic function translated horizontally 1 unit to the right.



[2 marks]

d This is a vertical line passing through $(-3, 0)$.



[1 mark]

e This is a parabola with x-intercepts when $y = 0$

$x^2 + 3x = 0$

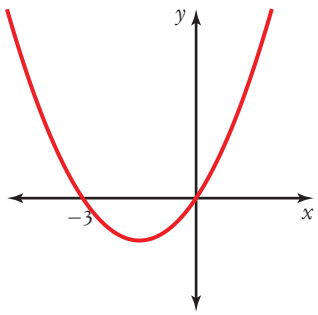
$x(x + 3) = 0$

$x = 0 \qquad x + 3 = 0$

$x = 0 \qquad x = -3$

For the y-intercept, $x = 0$

$f(0) = 0^2 + 3(0) = 0$



[2 marks]

2 a Substitute (1, 0) and (0, 4) into $m = \frac{y_2 - y_1}{x_2 - x_1}$ **[1 mark]**

$m = \frac{4-0}{0-1} = -4$ **[1 mark]**

b Substitute 60° into $m = \tan(\theta)$ **[1 mark]**

$m = \tan(60°) = \sqrt{3}$ **[1 mark]**

c $x - 5y - 3 = 0$

$x - 3 = 5y$

$\frac{1}{5}x - \frac{3}{5} = y$ **[1 mark]**

$m_1 = \frac{1}{5}$ **[1 mark]**

For perpendicular lines, $m_1 m_2 = -1$

$\therefore m_2 = -5$ **[1 mark]**

d Substitute $(-7, 6)$ and $(4, -3)$ into $m = \frac{y_2 - y_1}{x_2 - x_1}$ **[1 mark]**

$m = \frac{-3-6}{4-(-7)} = -\frac{9}{11}$ **[1 mark]**

3 a $y = 3x + 1$ has $m_1 = 3$

For parallel lines, $m_1 = m_2$

$\therefore m_2 = 3$ **[1 mark]**

Substitute $m_2 = 3$ and $(-1, 6)$ into $y - y_1 = m(x - x_1)$

$y - 6 = 3[x - (-1)]$

$= 3(x + 1)$

$= 3x + 3$

$y = 3x + 9$ **[1 mark]**

b General quadratic functions are in the form $y = ax^2 + bx + c$

Substitute (0, 1).

$1 = a(0)^2 + b(0) + c$

$1 = c$ **[1 mark]**

Substitute (3, 4).

$4 = a(3)^2 + b(3) + c$

$= 9a + 3b + 1$ (since $c = 1$)

$3 = 9a + 3b$

$1 = 3a + b$ [1] **[1 mark]**

Substitute $(1, -2)$.

$-2 = a(1)^2 + b(1) + c$

$= a + b + 1$ (since $c = 1$)

$-3 = a + b$ [2] **[1 mark]**

Solve simultaneous equations.

[1]–[2]:

$4 = 2a$

$2 = a$ **[1 mark]**

Substitute $a = 2$ into [1].

$1 = 3(2) + b$

$= 6 + b$

$-5 = b$

The function's equation is $y = 2x^2 - 5x + 1$ **[1 mark]**

c Substitute $(2, -2)$ and $r = 4$ into $(x - a)^2 + (y - b)^2 = r^2$

$(x - 2)^2 + (y - (-2))^2 = 4^2$ **[1 mark]**

$(x - 2)^2 + (y + 2)^2 = 16$

$x^2 - 4x + 4 + y^2 + 4y + 4 = 16$ **[1 mark]**

$x^2 - 4x + y^2 + 4y + 8 = 16$

$x^2 - 4x + y^2 + 4y - 8 = 0$ **[1 mark]**

4 a Use $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ to find the length of each side.

$AB = \sqrt{(2 - (-1))^2 + (4 - 0)^2}$

$= \sqrt{3^2 + 4^2}$

$= \sqrt{25}$

$= 5$ **[1 mark]**

$BC = 5$ (horizontal distance) **[1 mark]**

$CD = \sqrt{(7 - 4)^2 + (4 - 0)^2}$

$= \sqrt{3^2 + 4^2}$

$= \sqrt{25}$

$= 5$ **[1 mark]**

$AD = 5$ (horizontal distance) **[1 mark]**

Since all sides are equal, $ABCD$ is a rhombus. **[1 mark]**

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b Use $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ to find the midpoint of each diagonal.

Midpoint of $AC = \left(\frac{-1 + 7}{2}, \frac{0 + 4}{2}\right)$

$= (3, 2)$ **[1 mark]**

Midpoint of $BD = \left(\frac{2 + 4}{2}, \frac{4 + 0}{2}\right)$

$= (3, 2)$ **[1 mark]**

Since the midpoints are equal, the diagonals meet at the midpoint.

So the diagonals bisect each other. **[1 mark]**

5 a $\triangle = b^2 - 4ac$

$= 1^2 - 4(-3)(-4)$

$= -47$ **[1 mark]**

Since $\triangle < 0$, there are no real zeros. **[1 mark]**

b $x = -\frac{b}{2a}$

$= -\frac{1}{2 \times (-3)}$ **[1 mark]**

$= \frac{1}{6}$

The equation of the axis of symmetry is $x = \frac{1}{6}$ **[1 mark]**

c Substitute $x = \frac{1}{6}$.

$y = -3\left(\frac{1}{6}\right)^2 + \frac{1}{6} - 4$

$= -3\frac{11}{12}$

The turning point is $\left(\frac{1}{6}, -3\frac{11}{12}\right)$. **[1 mark]**

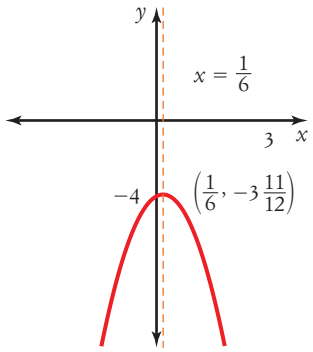
It is a maximum since $a < 0$ $(a = -3)$. **[1 mark]**

d No x-intercepts since $\triangle < 0$.

For the y-intercept, $x = 0$

$y = -3(0)^2 + 0 - 4$

$= -4$



[2 marks]

6 a For zeros, $P(x) = 0$

$a(x - b)(x - c)(x - d) = 0$ **[1 mark]**

$x - b = 0 \qquad x - c = 0 \qquad x - d = 0$

$x = b \qquad x = c \qquad x = d$

There are 3 zeros at $x = b, c, d$ **[1 mark]**

b The highest power of x is x^3 (if you expand the brackets). So $P(x)$ has degree 3. **[1 mark]**

c Monic: $a = 1$

Substitute $b = 2, c = -1, d = 4$

$P(x) = (x - 2)(x + 1)(x - 4)$ **[1 mark]**

Zeros are $x = 2, -1, 4$ (from part **a**).

For $x < -1$: $y < 0$

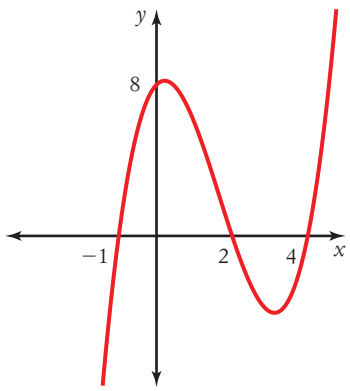
For $-1 < x < 2$: $y > 0$

For $2 < x < 4$: $y < 0$

For $x > 4$: $y > 0$

For the y-intercept, $x = 0$

$y = (0 - 2)(0 + 1)(0 - 4) = 8$



[2 marks]

7 a The graph is discontinuous at $x = \pm 3$ (the asymptotes) **[1 mark]**

b The graph is a function (vertical line test) **[1 mark]**

c i The gradient is positive (rising) for $\{x: x < -3, -3 < x < 0\}$ **[1 mark]**

ii The gradient is negative (falling) for $\{x: 0 < x < 3, x > 3\}$ **[1 mark]**

iii The graph is increasing (rising) for $\{x: x < -3, -3 < x < 0\}$ **[1 mark]**

iv The graph is decreasing (falling) for $\{x: 0 < x < 3, x > 3\}$ **[1 mark]**

Part C

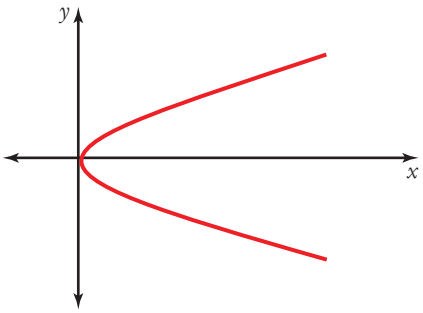
1 a $y^2 = 4ax$

$y = \pm\sqrt{4ax}$

$= \pm 2\sqrt{ax}$

The 2 functions are $y = 2\sqrt{ax}$ and $y = -2\sqrt{ax}$ **[1 mark]**

b $ax \geq 0$ so $x \geq 0$ $(a > 0)$



[2 marks]

c Substitute Q into the equation $y^2 = 4ax$

LHS $= y^2$

$= (2aq)^2$

$= 4a^2q^2$

RHS $= 4ax$

$= 4a(aq^2)$

$= 4a^2q^2$

Since LHS = RHS, point Q lies on $y^2 = 4ax$ **[1 mark]**

d Substitute P and Q into $m = \frac{y_2 - y_1}{x_2 - x_1}$

$$m = \frac{2ap - 2aq}{ap^2 - aq^2}$$

[1 mark]

$$= \frac{2a(p - q)}{a(p^2 - q^2)}$$

$$= \frac{2(p - q)}{p^2 - q^2}$$

$$= \frac{2(p - q)}{(p + q)(p - q)}$$

$$= \frac{2}{p + q}$$

[1 mark]

e $y - y_1 = m(x - x_1)$

$$y - 2aq = \frac{2}{p+q}(x - aq^2)$$

[1 mark]

$$(p + q)(y - 2aq) = 2(x - aq^2)$$

$$(p + q)y - (p + q)2aq = 2x - 2aq^2$$

$$(p + q)y - 2apq - 2aq^2 = 2x - 2aq^2$$

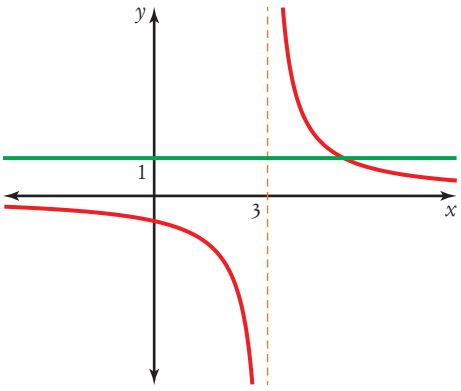
Moving all terms to the RHS and simplifying, we have

$$0 = 2x - (p + q)y + 2apq$$

[2 marks]

2 a $y = \frac{2}{x-3}$ is a hyperbola in the 1st and 3rd quadrants with asymptotes at $x = 3$ and $y = 0$.

$y = 1$ is a horizontal line with y-intercept 1.



[3 marks]

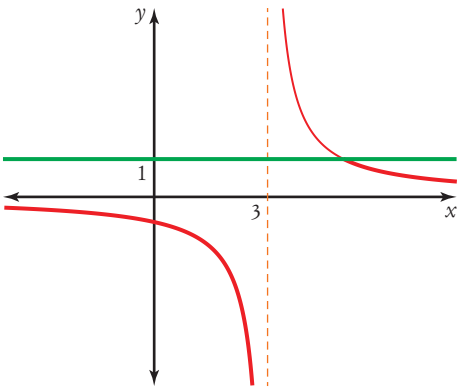
b Graphically, $\frac{2}{x-3} \leq 1$ is the part of the graph $y = \frac{2}{x-3}$ that is on and below the line $y = 1$.
Find the point of intersection of the graphs by substituting $y = 1$ into $y = \frac{2}{x-3}$.

$$1 = \frac{2}{x-3}$$

$$x - 3 = 2$$

$$x = 5$$

[1 mark]



The solution is $x < 3, x \geq 5$ **[1 mark]**

You could write this solution as $\frac{2}{x-3} \leq 1$ when $\{x < 3\} \cup \{x \geq 5\}$.